Field Effect Transistor - N-Channel, Enhancement Mode

BS170, MMBF170

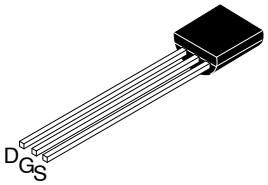
General Description

These N-Channel enhancement mode field effect transistors are produced using **onsemi**'s proprietary, high cell density, DMOS technology. These products have been designed to minimize on-state resistance while provide rugged, reliable, and fast switching performance. They can be used in most applications requiring up to 500 mA DC. These products are particularly suited for low voltage, low current applications such as small servo motor control, power MOSFET gate drivers, and other switching applications.

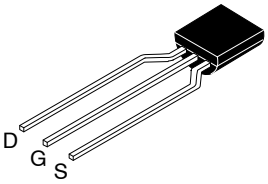
Features

- High Density Cell Design for Low $R_{DS(ON)}$
- Voltage Controlled Small Signal Switch
- Rugged and Reliable
- High Saturation Current Capability
- These are Pb-Free Devices

BS170

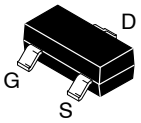


TO-92 3 4.825x4.76
CASE 135AN

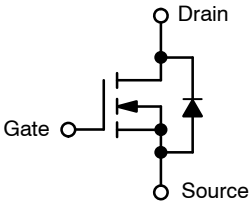


TO-92 3 4.83x4.76 LEADFORMED
CASE 135AR

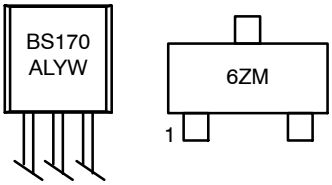
MMBF170



SOT-23
CASE 318-08



MARKING DIAGRAM



BS170, 6Z	= Device Code
A	= Assembly Plant Code
L	= Wafer Lot Number
YW	= Assembly Start Week
M	= Date Code

ORDERING INFORMATION

See detailed ordering and shipping information on page 8 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 8.

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ABSOLUTE MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

Symbol	Parameter		BS170	MMBF170	Unit
V_{DSS}	Drain-Source Voltage		60		V
V_{DGR}	Drain-Gate Voltage (R_{GS} ≤ 1 MΩ)		60		V
V_{GSS}	Gate-Source Voltage		±20		V
I_D	Drain Current	– Continuous	500	500	mA
		– Pulsed	1200	800	
T_J, T_{STG}	Operating and Storage Temperature Range		– 55 to 150		°C
T_L	Maximum Lead Temperature for Soldering Purposes, 1/16” from Case for 10 Seconds		300		°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)

Symbol	Parameter	BS170	MMBF170	Unit
P_D	Maximum Power Dissipation Derate above 25 °C	830 6.6	300 2.4	mW mW/°C
$R_{\theta JA}$	Thermal Resistance, Junction to Ambient	150	417	°C/W

ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)

Symbol	Parameter	Test Condition	Type	Min	Typ	Max	Unit
OFF CHARACTERISTICS							
BV_{DSS}	Drain-Source Breakdown Voltage	V_{GS} = 0 V, I_D = 100 μA	All	60	–	–	V
I_{DSS}	Zero Gate Voltage Drain Current	V_{DS} = 25 V, V_{GS} = 0 V	All	–	–	0.5	μA
I_{GSSF}	Gate - Body Leakage, Forward	V_{GS} = 15 V, V_{DS} = 0 V	All	–	–	10	nA
ON CHARACTERISTICS (Note 1)							
$V_{GS(th)}$	Gate Threshold Voltage	V_{DS} = V_{GS}, I_D = 1 mA	All	0.8	2.1	3	V
$R_{DS(ON)}$	Static Drain-Source On-Resistance	V_{GS} = 10 V, I_D = 200 mA	All	–	1.2	5	Ω
g_{FS}	Forward Transconductance	V_{DS} = 10 V, I_D = 200 mA	BS170	–	320	–	mS
		V_{DS} ≥ 2 $V_{DS(on)}$, I_D = 200 mA	MMBF170	–	320	–	
DYNAMIC CHARACTERISTICS							
C_{iss}	Input Capacitance	V_{DS} = 10 V, V_{GS} = 0 V, f = 1.0 MHz	All	–	24	40	pF
C_{oss}	Output Capacitance		All	–	17	30	pF
C_{rss}	Reverse Transfer Capacitance		All	–	7	10	pF
SWITCHING CHARACTERISTICS (Note 1)							
t_{on}	Turn-On Time	V_{DD} = 25 V, I_D = 200 mA, V_{GS} = 10 V, R_{GEN} = 25 Ω	BS170	–	–	10	ns
		V_{DD} = 25 V, I_D = 500 mA, V_{GS} = 10 V, R_{GEN} = 50 Ω	MMBF170	–	–	10	
t_{off}	Turn-Off Time	V_{DD} = 25 V, I_D = 200 mA, V_{GS} = 10 V, R_{GEN} = 25 Ω	BS170	–	–	10	ns
		V_{DD} = 25 V, I_D = 500 mA, V_{GS} = 10 V, R_{GEN} = 50 Ω	MMBF170	–	–	10	

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2.0%.

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TYPICAL ELECTRICAL CHARACTERISTICS

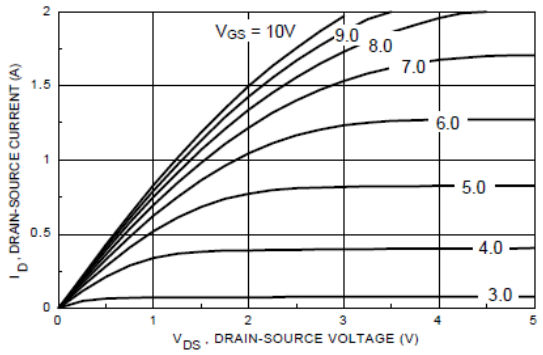

Figure 1. On-Region Characteristics

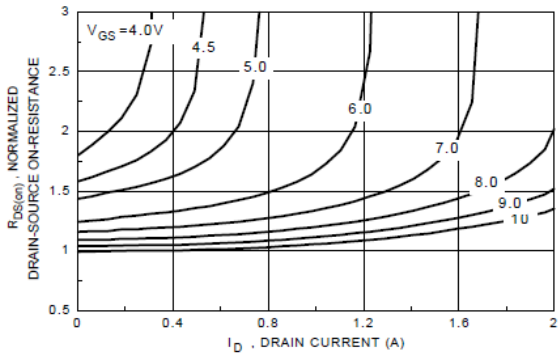

Figure 2. On-Resistance Variation with Gate Voltage and Drain Current

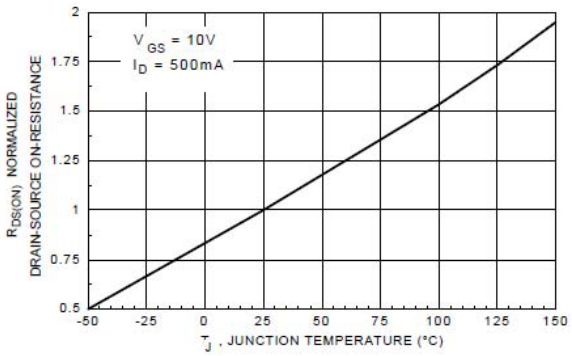

Figure 3. On-Resistance Variation with Temperature

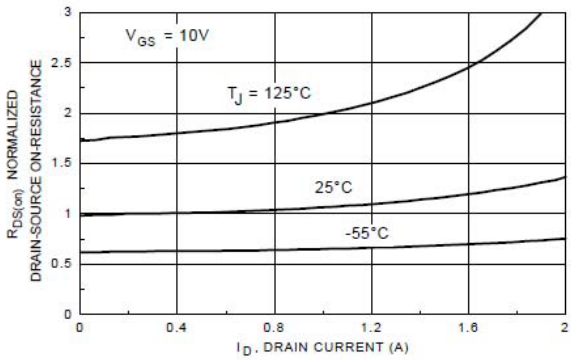

Figure 4. On-Resistance Variation with Drain Current and Temperature

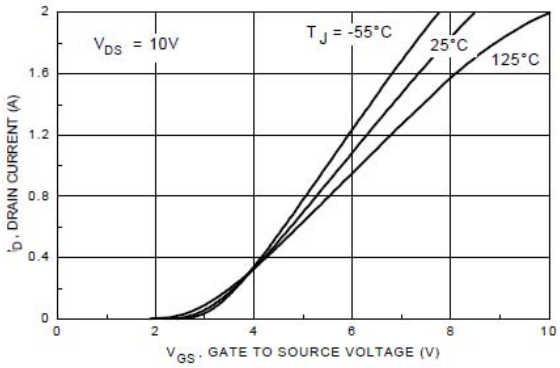

Figure 5. Transfer Characteristics

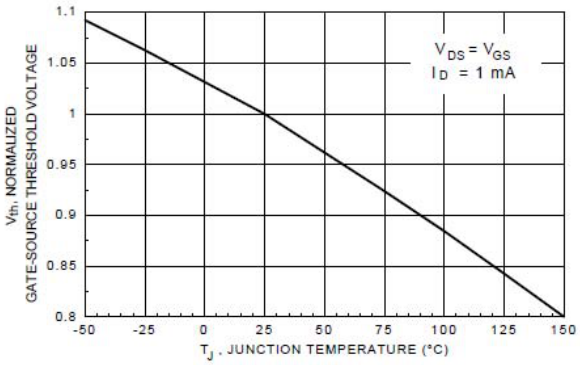

Figure 6. Gate Threshold Variation with Temperature

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TYPICAL ELECTRICAL CHARACTERISTICS (CONTINUED)

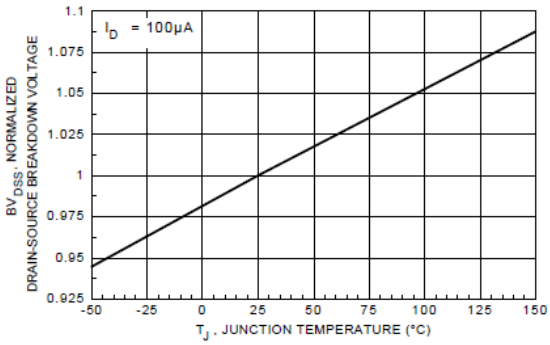

Figure 7. Breakdown Voltage Variation with Temperature

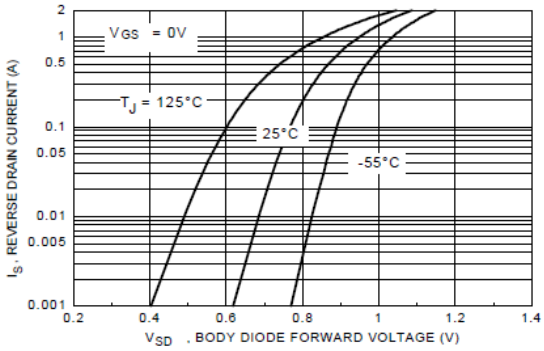

Figure 8. Body Diode Forward Voltage Variation with Current and Temperature

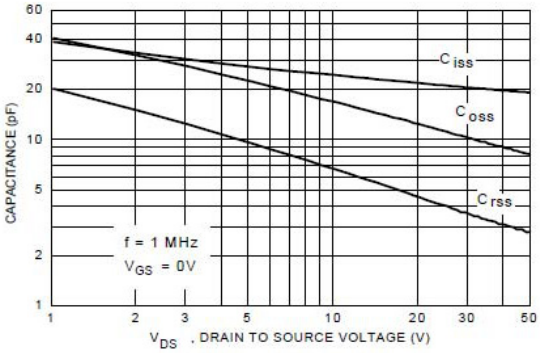

Figure 9. Capacitance Characteristics

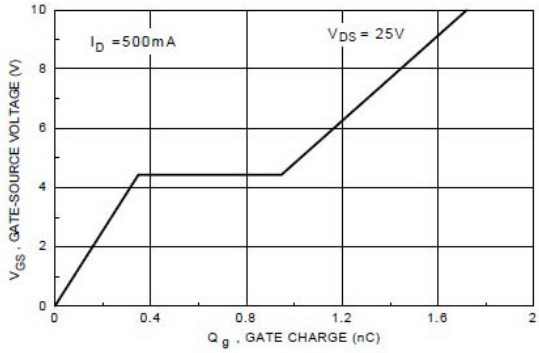

Figure 10. Gate Charge Characteristics

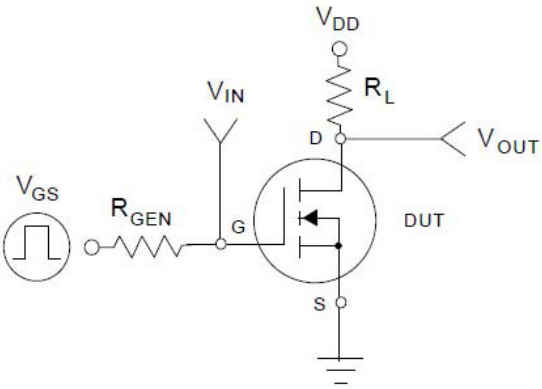

Figure 11. Switching Test Circuit

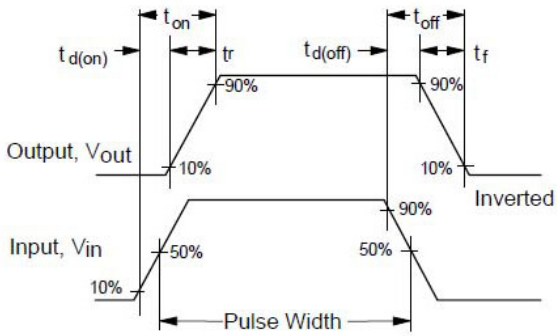

Figure 12. Switching Waveforms

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TYPICAL ELECTRICAL CHARACTERISTICS (CONTINUED)

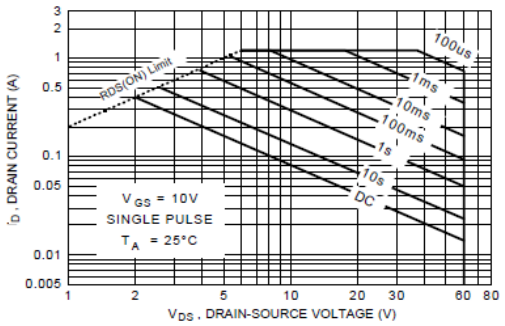

Figure 13. BS170 Maximum Safe Operating Area

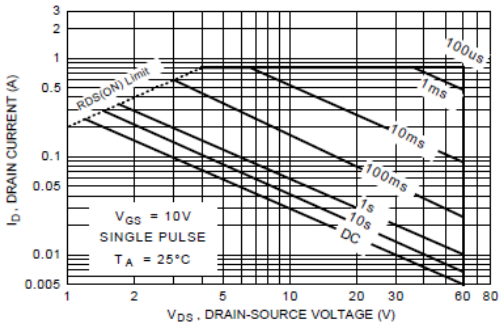

Figure 14. MMBF170 Maximum Safe Operating Area

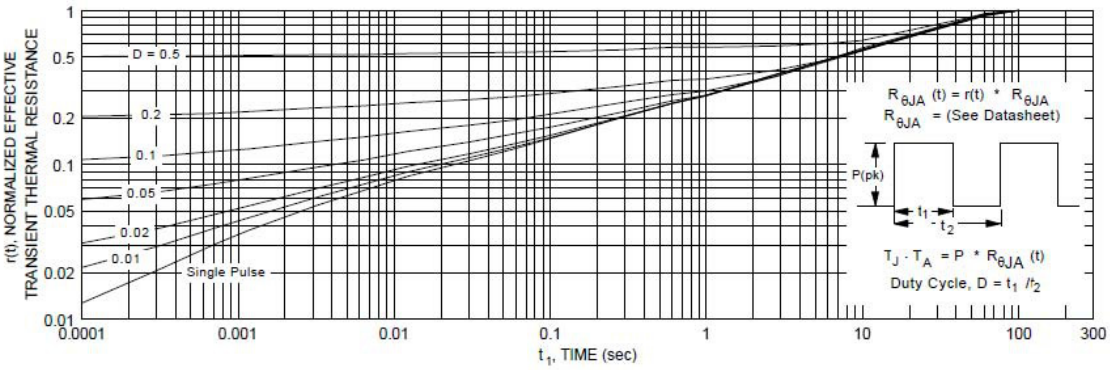

Figure 15. TO-92, BS170 Transient Thermal Response Curve

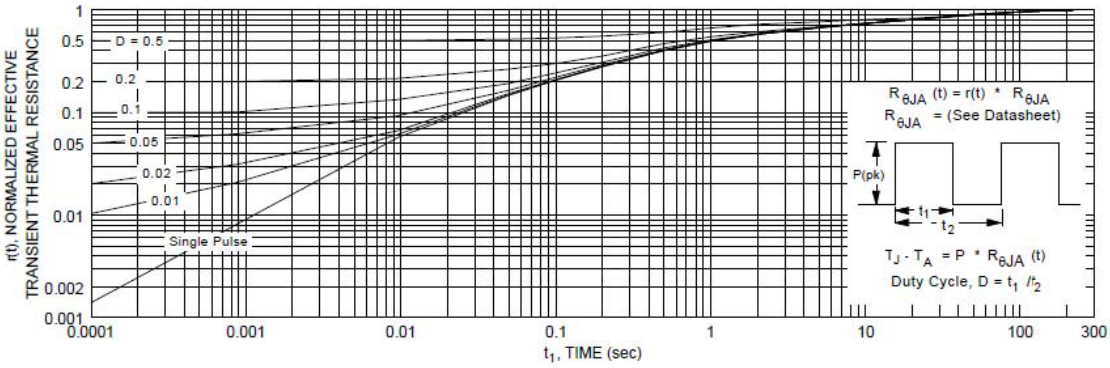

Figure 16. SOT-23, MMBF170 Transient Thermal Response Curve

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ORDERING INFORMATION

Part Number	Package	Lead Frame	Pin Array	Shipping†
BS170	TO-92 (Pb-Free)	Straight	D G S	10,000 Units / Bulk
BS170-D26Z	TO-92 (Pb-Free)	Forming	D G S	2,000 / Tape & Reel
BS170-D27Z	TO-92 (Pb-Free)	Forming	D G S	2,000 / Tape & Reel
MMBF170	SOT-23 (Pb-Free)			3,000 / Tape & Reel

DISCONTINUED (Note 11)

BS170-D74Z	TO-92 (Pb-Free)	Forming	D G S	2,000 / Ammo
BS170-D75Z	TO-92 (Pb-Free)	Forming	D G S	2,000 / Ammo

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

2. **DISCONTINUED:** These devices are not available. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.

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REVISION HISTORY

Revision	Description of Changes	Date
9	BS170-D75Z OPN Marked as Discontinued.	02/12/2026

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

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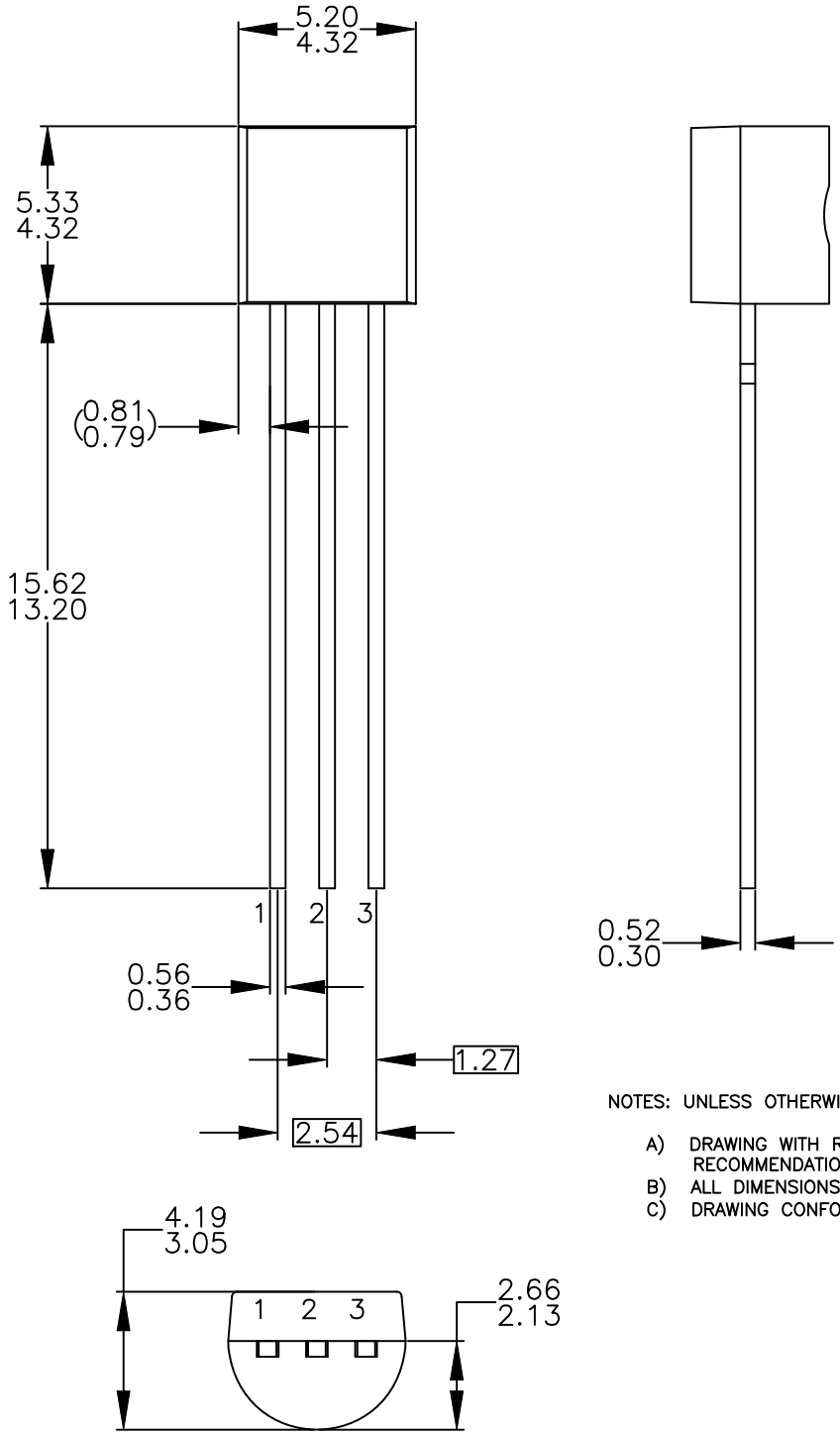
TO−92 3 4.825x4.76
CASE 135AN
ISSUE O

DATE 31 JUL 2016



NOTES: UNLESS OTHERWISE SPECIFIED

A) DRAWING WITH REFERENCE TO JEDEC TO−92 RECOMMENDATIONS.
B) ALL DIMENSIONS ARE IN MILLIMETERS.
C) DRAWING CONFORMS TO ASME Y14.5M−2009.

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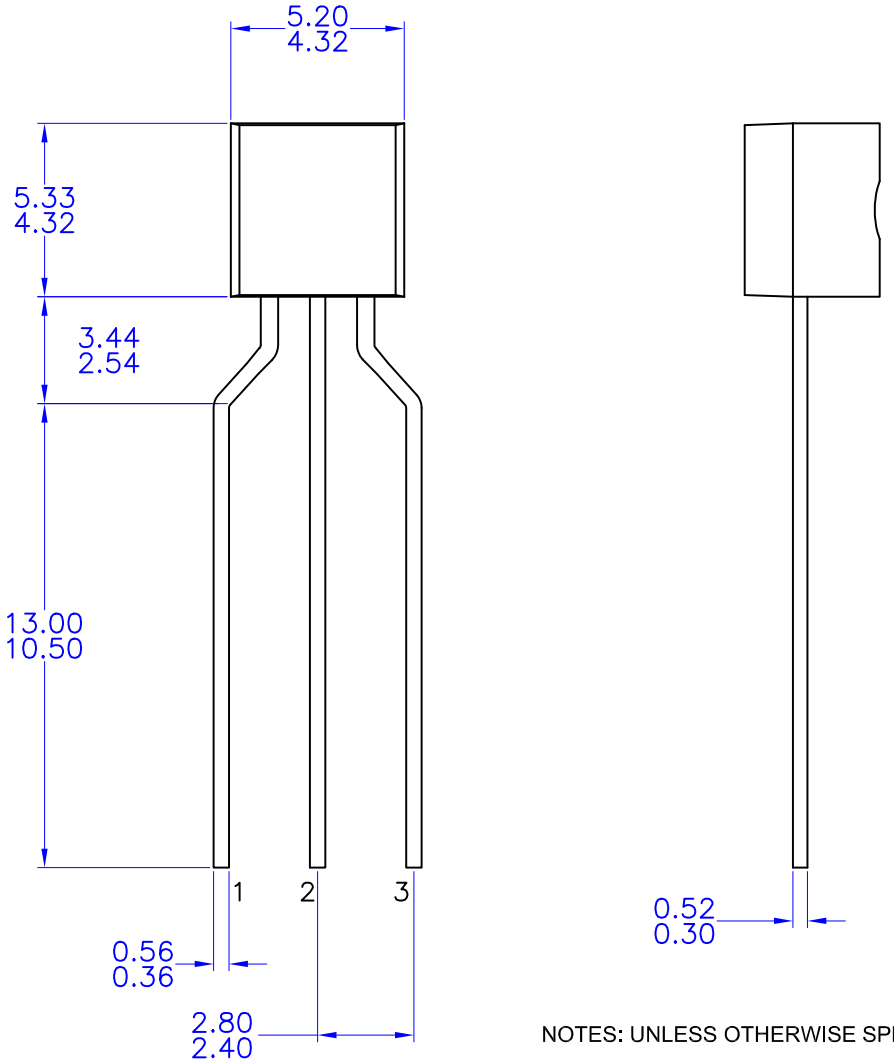
TO−92 3 4.83x4.76 LEADFORMED
CASE 135AR
ISSUE O

DATE 30 SEP 2016



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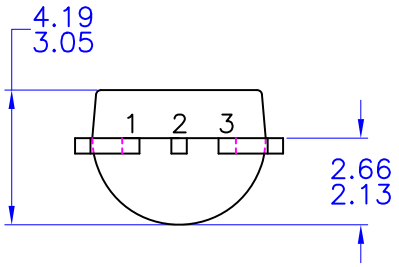

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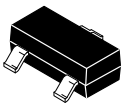
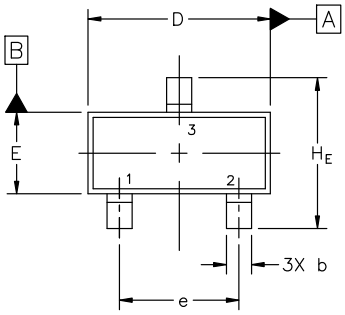


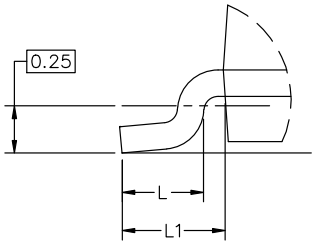
SCALE 4:1

SOT−23 (TO−236) 2.90x1.30x1.00 1.90P
CASE 318
ISSUE AU

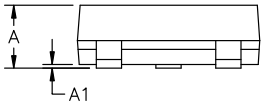
DATE 14 AUG 2024



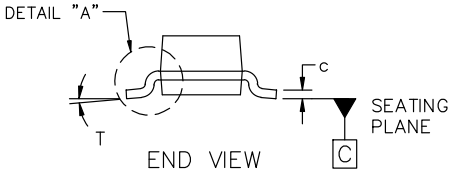
TOP VIEW



DETAIL "A"
Scale 3:1



SIDE VIEW

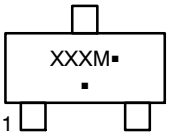


END VIEW

MILLIMETERS			
DIM	MIN	NOM	MAX
A	0.89	1.00	1.11
A1	0.01	0.06	0.10
b	0.37	0.44	0.50
c	0.08	0.14	0.20
D	2.80	2.90	3.04
E	1.20	1.30	1.40
e	1.78	1.90	2.04
L	0.30	0.43	0.55
L1	0.35	0.54	0.69
H_E	2.10	2.40	2.64
T	0°	−−−	10°

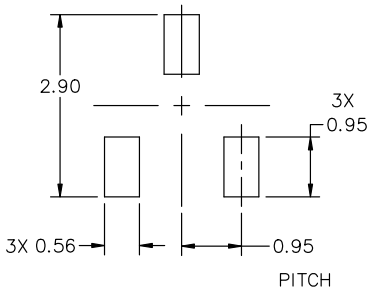
NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSIONS: MILLIMETERS.
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF THE BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.

GENERIC MARKING DIAGRAM*



XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.



RECOMMENDED
MOUNTING FOOTPRINT

* For additional information on our Pb−Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

STYLES ON PAGE 2

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ISSUE AU

DATE 14 AUG 2024

STYLE 1 THRU 5:
CANCELLED

STYLE 6:
PIN 1. BASE
2. EMITTER
3. COLLECTOR

STYLE 7:
PIN 1. EMITTER
2. BASE
3. COLLECTOR

STYLE 8:
PIN 1. ANODE
2. NO CONNECTION
3. CATHODE

STYLE 9:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 10:
PIN 1. DRAIN
2. SOURCE
3. GATE

STYLE 11:
PIN 1. ANODE
2. CATHODE
3. CATHODE−ANODE

STYLE 12:
PIN 1. CATHODE
2. CATHODE
3. ANODE

STYLE 13:
PIN 1. SOURCE
2. DRAIN
3. GATE

STYLE 14:
PIN 1. CATHODE
2. GATE
3. ANODE

STYLE 15:
PIN 1. GATE
2. CATHODE
3. ANODE

STYLE 16:
PIN 1. ANODE
2. CATHODE
3. CATHODE

STYLE 17:
PIN 1. NO CONNECTION
2. ANODE
3. CATHODE

STYLE 18:
PIN 1. NO CONNECTION
2. CATHODE
3. ANODE

STYLE 19:
PIN 1. CATHODE
2. ANODE
3. CATHODE−ANODE

STYLE 20:
PIN 1. CATHODE
2. ANODE
3. GATE

STYLE 21:
PIN 1. GATE
2. SOURCE
3. DRAIN

STYLE 22:
PIN 1. RETURN
2. OUTPUT
3. INPUT

STYLE 23:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 24:
PIN 1. GATE
2. DRAIN
3. SOURCE

STYLE 25:
PIN 1. ANODE
2. CATHODE
3. GATE

STYLE 26:
PIN 1. CATHODE
2. ANODE
3. NO CONNECTION

STYLE 27:
PIN 1. CATHODE
2. CATHODE
3. CATHODE

STYLE 28:
PIN 1. ANODE
2. ANODE
3. ANODE

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